FUNCTIONAL TESTING DOCUMENT

Website: https://practicesoftwaretesting.com/

Requirements-Based Testing | RTM | Playwright | JUnit

1. Purpose and Scope

This document derives functional test cases from the SRS for the Practice Software Testing Toolshop. It contains the requirements baseline, detailed test cases with execution placeholders, a Requirements Traceability Matrix (RTM), selected Playwright screenshot evidence placeholders, and a JUnit testing plan. The SRS is the primary authority for requirement IDs and intended behavior.

2. Requirements

Table 1 lists the functional requirements reproduced from Section 7 of the SRS. Requirement descriptions are retained from the source; verification notes are included to preserve the SRS acceptance guidance.

Requirement ID	Module	Requirement Description	Verification Notes / Acceptance Criteria
PROD-001	Product Browsing	The system shall display the primary product catalog grid on the home page with product images, titles, brand names, and prices.	Verify layout on desktop and mobile viewports. Check image rendering.
PROD-002	Product Browsing	The system shall clearly display an 'Out of Stock' badge for items with zero inventory and disable the add-to-cart button.	Verify on website whether button is disabled or hidden when out of stock.
PROD-003	Product Browsing	The system shall support pagination or dynamic scroll loading when the product catalog exceeds the per-page display limit.	Verify pagination buttons (Next, Previous, Page Numbers) or infinite scroll.
PROD-004	Product Browsing	The system shall allow users to click any product card to navigate directly to its dedicated Product Details view.	Verify URL updates to '/product/{id}' and correct item data is loaded.
SRCH-001	Search & Filters	The system shall filter displayed products dynamically when a search keyword is entered into the search input box.	Verify search with valid product names (e.g., 'Hammer', 'Pliers', 'Drill').
SRCH-002	Search & Filters	The system shall filter products based on selected category checkboxes (e.g., Hand Tools, Power Tools).	Verify multi-category selection and category deselection behaviors.
SRCH-003	Search & Filters	The system shall filter products by brand checkboxes (e.g., ForgeFlex, ProTool) matching selected criteria.	Verify brand filtering independently and combined with category filters.
SRCH-004	Search & Filters	The system shall dynamically filter products based on the minimum and maximum values of the price range slider.	Verify slider edge values and that catalog updates without full page reload.
SRCH-005	Search & Filters	The system shall sort catalog items according to user selection: Name (A-Z, Z-A) and Price (Low to High, High to Low).	Verify alphabetical and numerical ordering accuracy of returned items.
SRCH-006	Search & Filters	The system shall display a clear, user-friendly 'No products found' message when no items match search/filter criteria.	Verify with non-existent keywords (e.g., 'XYZ999INVALID') and restrictive sliders.
DET-001	Product Details	The system shall render complete product specifications, including title, category, brand, description, and price.	Verify data consistency between catalog card and detail page view.
DET-002	Product Details	The system shall allow users to adjust item quantity using '+' and '-' buttons or direct numeric input (minimum value: 1).	Verify boundary conditions: quantity = 0, negative numbers, and exceeding stock.
DET-003	Product Details	The system shall add the selected quantity of the product to the shopping cart when the 'Add to Cart' button is clicked.	Verify cart count badge increments and confirmation toast message appears.
DET-004	Product Details	The system shall prevent adding quantities that exceed the currently available warehouse stock level.	Verify error message or input restriction when requested quantity > stock.
USER-001	User Account	The system shall provide a registration form requiring Name, Address, Email, and Password with validation.	Verify mandatory field validation, email format regex, and password rules.
USER-002	User Account	The system shall prevent registration with an email address that already exists in the customer database.	Verify error feedback: 'A customer with this email address already exists.'
USER-003	User Account	The system shall authenticate registered customers with valid email and password credentials.	Verify successful login redirects to user account dashboard or home page.
USER-004	User Account	The system shall reject authentication attempts with invalid passwords or non-registered email addresses.	Verify security-compliant error message: 'Invalid email or password.'
USER-005	User Account	The system shall allow authenticated customers to view and update their profile details and shipping address.	Verify changes are persisted across subsequent sessions and database re-queries.
USER-006	User Account	The system shall provide a secure Logout mechanism that	Verify logout redirects to home/login

		clears session storage and terminates active tokens.	and restricted routes cannot be accessed.
CART-001	Shopping Cart	The system shall display all added products with thumbnail, title, unit price, quantity, line total, and grand total.	Verify accurate mathematical calculation: Line Total = Unit Price * Quantity.
CART-002	Shopping Cart	The system shall permit modification of item quantities within the cart and dynamically update line subtotals and grand total.	Verify immediate subtotal re-calculation without manual page refresh.
CART-003	Shopping Cart	The system shall allow users to remove an item from the cart via a delete/trash icon.	Verify item is removed from view, cart count decrements, and total updates.
CART-004	Shopping Cart	The system shall retain cart items across browser sessions using local storage or user session persistence.	Verify cart contents remain intact after page reload or reopening browser tab.
CART-005	Shopping Cart	The system shall display an empty cart message with a 'Return to store' button when all items are removed.	Verify UI state when cart item count equals zero.
CHK-001	Checkout Flow	The system shall require the user to be authenticated before proceeding to the shipping and payment checkout steps.	Verify unauthenticated user is redirected to Login/Register screen.
CHK-002	Checkout Flow	The system shall capture and validate shipping address information (Address, City, State, Postal Code, Country).	Verify inline validation alerts for missing mandatory delivery fields.
CHK-003	Checkout Flow	The system shall allow selection of supported payment methods (Credit Card, Buy Now Pay Later, Cash on Delivery).	Verify payment-specific input fields appear (e.g., Card Number, Expiry, CVV).
CHK-004	Checkout Flow	The system shall validate credit card format, expiration date, and CVV before authorizing order submission.	Verify rejection of expired cards, invalid card lengths, and letters in CVV.
CHK-005	Checkout Flow	The system shall generate a confirmed order invoice with a unique Order Reference ID upon transaction completion.	Verify confirmation view, invoice download (if supported), and cart reset.
CHK-006	Checkout Flow	The system shall deduct purchased product quantities from inventory upon successful order completion.	Verify stock level decrements on catalog/detail page (Verify on website).

3. Test Cases Matrix

The SRS provides 34 suggested scenarios (TC-SCN-001 to TC-SCN-034). The cases below use these scenarios as the baseline and provide execution fields. Actual Result and Status are placeholders for the tester to complete after execution.

Test Case ID	Requirement ID(s)	Level	Test Scenario / Steps Summary	Expected Result	Actual Result	Status
TC-SCN-001	PROD-001, PROD-004	Product Browsing	Verify default catalog loads with valid product cards, images, prices, and title text.	Verify against SRS and live website. Record data used in execution.	Website loads with catalog and product cards	PASS
TC-SCN-002	PROD-001	Product Browsing	Verify system handles broken image links or missing product thumbnails gracefully with placeholder images.	Verify against SRS and live website. Record data used in execution.	Product catalogue heading and product links remained visible when image requests were blocked.	PASS
TC-SCN-003	PROD-001, PROD-003, PROD-004	Product Browsing	Verify catalog pagination navigates accurately to subsequent pages and updates URL parameters.	Verify against SRS and live website. Record data used in execution.	Clicking Page 2 completed without a test error and pagination actually changed	PASS
TC-SCN-004	PROD-001, PROD-003	Product Browsing	Verify direct URL navigation to an out-of-range pagination number (e.g., page=99999) displays empty catalog gracefully.	Verify against SRS and live website. Record data used in execution.	Opening the catalogue with ?page=99999 displayed 9 products instead of an empty catalogue.	FAIL
TC-SCN-005	SRCH-001	Search & Filters	Verify searching for exact matching product keyword (e.g., 'Pliers') returns all relevant items.	Verify against SRS and live website. Record data used in execution.	Search results included the expected products, including Combination Pliers and Long Nose Pliers.	PASS
TC-SCN-006	SRCH-001, SRCH-006	Search & Filters	Verify search input with random alphanumeric string (e.g., 'ZXQ987!#@') returns 'No products found' banner.	Verify against SRS and live website. Record data used in execution.	Searching for ZXQ987!#@ displayed “No products found” and no product links.	PASS
TC-SCN-007	PROD-001	Search & Filters	Verify submitting an empty search query or whitespace-only input maintains current catalog without crashing.	Verify against SRS and live website. Record data used in execution.	Submitting an empty search preserved the product catalogue and its product count.	PASS
TC-SCN-008	SRCH-002, SRCH-003	Search & Filters	Verify multi-faceted filtering: selecting category 'Power Tools' + Brand 'ForgeFlex' + Price < $100.	Verify against SRS and live website. Record data used in execution.	The selected Power Tools category, ForgeFlex Tools brand, and price constraint produced results satisfying the test's price assertion.	PASS
TC-SCN-009	SRCH-004	Search & Filters	Verify setting price slider min value greater than max value (if permitted by UI) or restrictive bounds yielding 0 results.	Verify against SRS and live website. Record data used in execution.	Setting the maximum-price slider to its minimum resulted in zero product links.	PASS
TC-SCN-010	SRCH-005	Search & Filters	Verify sorting by 'Price: Low to High' arranges items in strict ascending numerical order.	Verify against SRS and live website. Record data used in execution.	Selecting Price (Low - High) produced product prices in ascending order.	PASS

TC-SCN-011	PROD-004, DET-001	Product Details	Verify selecting an item loads accurate title, description, price, stock status, and image.	Verify against SRS and live website. Record data used in execution.	Selecting a product opens its details page, displaying the correct product title and price.	PASS
TC-SCN-012	PROD-004, SRCH-006, DET-001	Product Details	Verify direct URL navigation to a non-existent product ID (e.g., '/product/invalid-id-999') renders 404 error page.	Verify against SRS and live website. Record data used in execution.	Navigating to an invalid product URL displays the “Related products” heading without product details or an explicit error message.	FAIL
TC-SCN-013	PROD-004, DET-001, DET-002	Product Details	Verify entering non-numeric characters (e.g., 'abc', '-5', '0') into quantity field is prevented or sanitized.	Verify against SRS and live website. Record data used in execution.	The quantity field restricts invalid quantities, and the application displays a warning when the maximum permitted quantity is exceeded.	PASS
TC-SCN-014	PROD-004, DET-001, DET-002, DET-004	Product Details	Verify entering quantity exceeding available stock (e.g., 9999 units) displays warning and restricts addition.	Verify against SRS and live website. Record data used in execution.	Entering a quantity of 9999 triggers a warning and restricts the quantity to 99.	PASS
TC-SCN-015	PROD-004, DET-001, DET-002, DET-003	Product Details	Verify clicking 'Add to Cart' with valid quantity updates cart count badge immediately.	Verify against SRS and live website. Record data used in execution.	Adding a product to the cart with quantity 1 succeeds, and the cart count updates to 1.	PASS
TC-SCN-016	USER-001	User Account	Verify user registration with all mandatory fields populated with valid data succeeds.	Verify against SRS and live website. Record data used in execution.	Registration with valid details was unsuccessful because the application flagged the supplied password as appearing in a data leak.	FAIL
TC-SCN-017	USER-001	User Account	Verify registration form submission with empty mandatory fields triggers explicit validation error alerts.	Verify against SRS and live website. Record data used in execution.	Submitting the empty registration form displays red “Value required” validation messages for mandatory fields.	PASS
TC-SCN-018	USER-001	User Account	Verify registration with malformed email format (e.g., 'user@com', 'plainaddress') is rejected.	Verify against SRS and live website. Record data used in execution.	The application displays an email-format validation message for the malformed address plainaddress.	PASS
TC-SCN-019	USER-001	User Account	Verify registration with an already registered email address displays duplicate account error.	Verify against SRS and live website. Record data used in execution.	Registering with the existing email admin@practicesoftwaretesting.com displays an “already exists” error and prevents duplicate registration.	PASS
TC-SCN-020	USER-001	User Account	Verify registration with password below required complexity/length triggers password policy warning.	Verify against SRS and live website. Record data used in execution.	Entering the weak password abc displays the password policy requirements and weak-password feedback.	PASS
TC-SCN-021	USER-006	User Account	Verify successful customer login with registered email and password generates valid session.	Verify against SRS and live website. Record data used in execution.	Login with valid demo credentials succeeded and allowed access beyond the login page.	PASS
TC-SCN-022	USER-004	User Account	Verify login failure with incorrect password displays 'Invalid email or password' without revealing which is wrong.	Verify against SRS and live website. Record data used in execution.	Login with an incorrect password was tested to verify that authentication fails and a generic error is displayed.	PASS
TC-SCN-023	USER-001	User Account	Verify SQL injection string in login fields (e.g., admin' OR '1'='1) is safely handled and rejected.	Verify against SRS and live website. Record data used in execution.	SQL-injection-like strings in the login fields were rejected without granting access.	PASS
TC-SCN-024	CART-001, CART-002	Shopping Cart	Verify adding multiple distinct products calculates correct individual line subtotals and aggregate grand total.	Verify against SRS and live website. Record data used in execution.	Multiple products were added to the cart and the individual subtotals and overall total were checked.	PASS
TC-SCN-025	DET-002, CART-001, CART-002	Shopping Cart	Verify modifying item quantity within cart dynamically re-calculates subtotal and grand total.	Verify against SRS and live website. Record data used in execution.	Modifying the cart quantity was tested to verify that the quantity and totals update accordingly.	PASS
TC-SCN-026	DET-002, CART-002	Shopping Cart	Verify updating cart quantity to zero or negative prompts item removal or restricts value to minimum 1.	Verify against SRS and live website. Record data used in execution.	Zero and negative quantities were tested; the cart quantity was constrained to a valid minimum.	PASS
TC-SCN-027	CART-001	Shopping Cart	Verify clicking delete icon removes product line item from cart and recalculates totals.	Verify against SRS and live website. Record data used in execution.	Removing a product from the cart removed its entry from the cart display.	PASS
TC-SCN-028	CART-004	Shopping Cart	Verify cart items persist across browser page refreshes and browser restart.	Verify against SRS and live website. Record data used in execution.	The product remained in the cart after refreshing the checkout page.	PASS
TC-SCN-029	CHK-005	Checkout Flow	Verify complete checkout workflow from Cart -> Address -> Payment -> Order Confirmation using valid test data.	Verify against SRS and live website. Record data used in execution.	Checkout reached the billing-address stage, but the street, city, and state fields did not reliably retain the intended values.	FAIL

TC-SCN-030	USER-003	Checkout Flow	Verify checkout progression is blocked if user is unauthenticated, redirecting to login.	Verify against SRS and live website. Record data used in execution.	The sign-in tab displayed the login form when selected during checkout without an authenticated session.	PASS
TC-SCN-031	USER-001, CHK-002	Checkout Flow	Verify submitting empty shipping address fields triggers mandatory field validation messages.	Verify against SRS and live website. Record data used in execution.	The application validates the billing address and prevents checkout from proceeding when the required shipping/billing address fields are empty.	PASS
TC-SCN-032	CHK-004	Checkout Flow	Verify entering expired credit card (e.g., month/year in the past) rejects payment authorization.	Verify against SRS and live website. Record data used in execution.	The application rejects an expired credit card expiration date and displays the validation message: "Expiration date must be in the future."	PASS
TC-SCN-033	CHK-004	Checkout Flow	Verify entering invalid CVV (e.g., 2 digits or letters) is rejected by client-side regex validation.	Verify against SRS and live website. Record data used in execution.	After selecting Credit Card, the application rejects a CVV containing only two digits and displays "CVV must be 3 or 4 digits." Valid card details are populated and validated through the payment form.	PASS
TC-SCN-034	CHK-005	Checkout Flow	Verify successful payment generates unique Order Reference ID and clears shopping cart.	Verify against SRS and live website. Record data used in execution.	After valid payment details are submitted, the application displays "Payment was successful." Clicking Confirm generates an order confirmation with an invoice number in the format INV- followed by digits, and the cart is cleared.	PASS

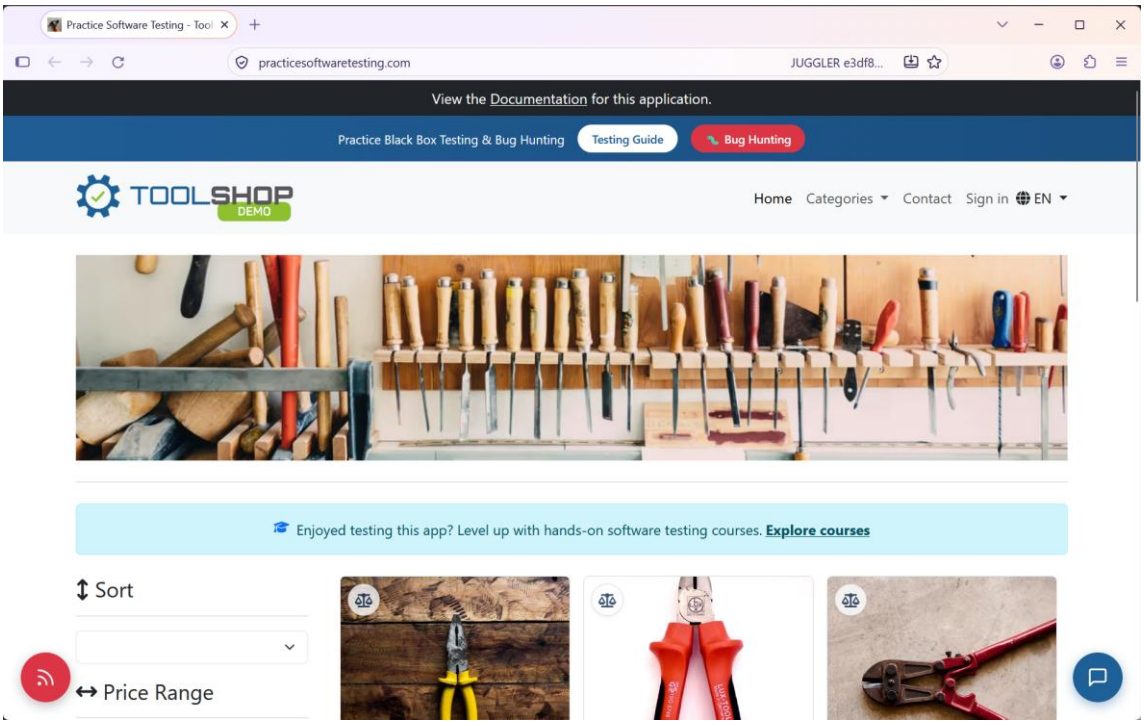
4. Requirements Traceability Matrix (RTM)

Requirement ID	Module	Requirement Summary	Mapped Test Case ID(s)
PROD-001	Product Browsing	The system shall display the primary product catalog grid on the home page with product images, titles, brand names, and prices.	TC-SCN-001, TC-SCN-002, TC-SCN-003, TC-SCN-004, TC-SCN-007
PROD-002	Product Browsing	The system shall clearly display an 'Out of Stock' badge for items with zero inventory and disable the add-to-cart button.	Additional test required
PROD-003	Product Browsing	The system shall support pagination or dynamic scroll loading when the product catalog exceeds the per-page display limit.	TC-SCN-003, TC-SCN-004
PROD-004	Product Browsing	The system shall allow users to click any product card to navigate directly to its dedicated Product Details view.	TC-SCN-001, TC-SCN-003, TC-SCN-011, TC-SCN-012, TC-SCN-013, TC-SCN-014, TC-SCN-015
SRCH-001	Search & Filters	The system shall filter displayed products dynamically when a search keyword is entered into the search input box.	TC-SCN-005, TC-SCN-006
SRCH-002	Search & Filters	The system shall filter products based on selected category checkboxes (e.g., Hand Tools, Power Tools).	TC-SCN-008
SRCH-003	Search & Filters	The system shall filter products by brand checkboxes (e.g., ForgeFlex, ProTool) matching selected criteria.	TC-SCN-008
SRCH-004	Search & Filters	The system shall dynamically filter products based on the minimum and maximum values of the price range slider.	TC-SCN-009
SRCH-005	Search & Filters	The system shall sort catalog items according to user selection: Name (A-Z, Z-A) and Price (Low to High, High to Low).	TC-SCN-010
SRCH-006	Search & Filters	The system shall display a clear, user-friendly 'No products found' message when no items match search/filter criteria.	TC-SCN-006, TC-SCN-012
DET-001	Product Details	The system shall render complete product specifications, including title, category, brand, description, and price.	TC-SCN-011, TC-SCN-012, TC-SCN-013, TC-SCN-014, TC-SCN-015
DET-002	Product Details	The system shall allow users to adjust item quantity using '+' and '-' buttons or direct numeric input (minimum value: 1).	TC-SCN-013, TC-SCN-014, TC-SCN-015, TC-SCN-025, TC-SCN-026
DET-003	Product Details	The system shall add the selected quantity of the product to the shopping cart when the 'Add to Cart' button is clicked.	TC-SCN-015
DET-004	Product Details	The system shall prevent adding quantities that exceed the currently available warehouse stock level.	TC-SCN-014
USER-001	User Account	The system shall provide a registration form requiring Name, Address, Email, and Password with validation.	TC-SCN-016, TC-SCN-017, TC-SCN-018, TC-SCN-019, TC-SCN-020, TC-SCN-023, TC-SCN-031

USER-002	User Account	The system shall prevent registration with an email address that already exists in the customer database.	Additional test required
USER-003	User Account	The system shall authenticate registered customers with valid email and password credentials.	TC-SCN-030
USER-004	User Account	The system shall reject authentication attempts with invalid passwords or non-registered email addresses.	TC-SCN-022
USER-005	User Account	The system shall allow authenticated customers to view and update their profile details and shipping address.	Additional test required
USER-006	User Account	The system shall provide a secure Logout mechanism that clears session storage and terminates active tokens.	TC-SCN-021
CART-001	Shopping Cart	The system shall display all added products with thumbnail, title, unit price, quantity, line total, and grand total.	TC-SCN-024, TC-SCN-025, TC-SCN-027
CART-002	Shopping Cart	The system shall permit modification of item quantities within the cart and dynamically update line subtotals and grand total.	TC-SCN-024, TC-SCN-025, TC-SCN-026
CART-003	Shopping Cart	The system shall allow users to remove an item from the cart via a delete/trash icon.	Additional test required
CART-004	Shopping Cart	The system shall retain cart items across browser sessions using local storage or user session persistence.	TC-SCN-028
CART-005	Shopping Cart	The system shall display an empty cart message with a 'Return to store' button when all items are removed.	Additional test required
CHK-001	Checkout Flow	The system shall require the user to be authenticated before proceeding to the shipping and payment checkout steps.	Additional test required
CHK-002	Checkout Flow	The system shall capture and validate shipping address information (Address, City, State, Postal Code, Country).	TC-SCN-031
CHK-003	Checkout Flow	The system shall allow selection of supported payment methods (Credit Card, Buy Now Pay Later, Cash on Delivery).	Additional test required
CHK-004	Checkout Flow	The system shall validate credit card format, expiration date, and CVV before authorizing order submission.	TC-SCN-032, TC-SCN-033
CHK-005	Checkout Flow	The system shall generate a confirmed order invoice with a unique Order Reference ID upon transaction completion.	TC-SCN-029, TC-SCN-034
CHK-006	Checkout Flow	The system shall deduct purchased product quantities from inventory upon successful order completion.	Additional test required

5. Playwright Automated Testing Evidence

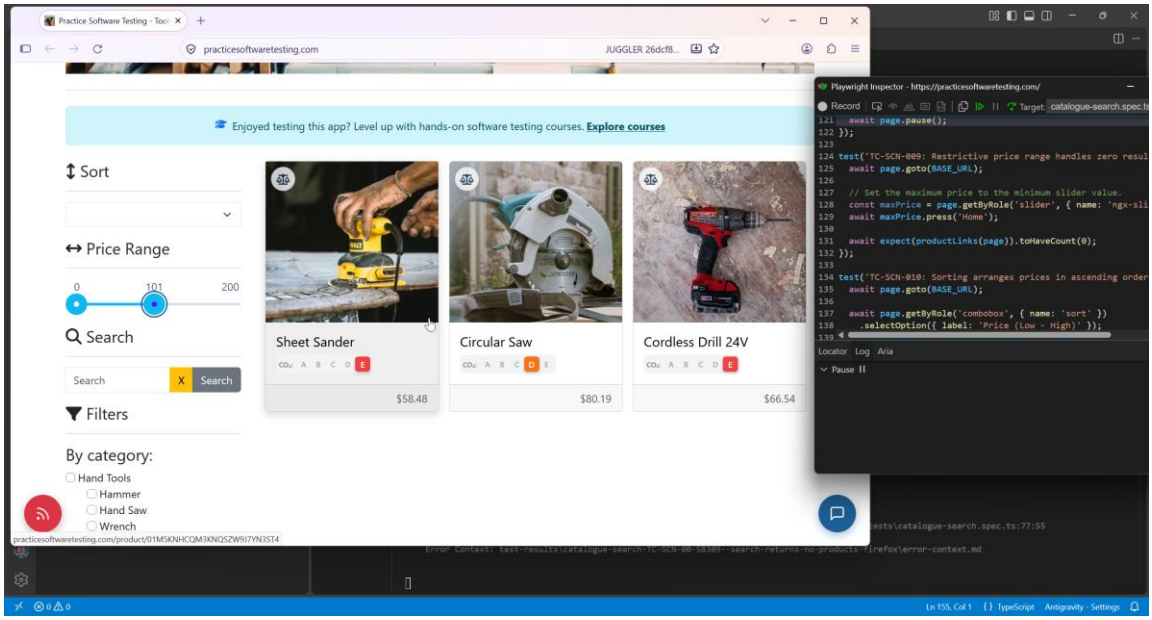
Figure 1 – Product catalogue loads



Playwright evidence for successful product catalogue rendering (TC-SCN-001) Captured in [Firefox Nightly(playwright-browser).]

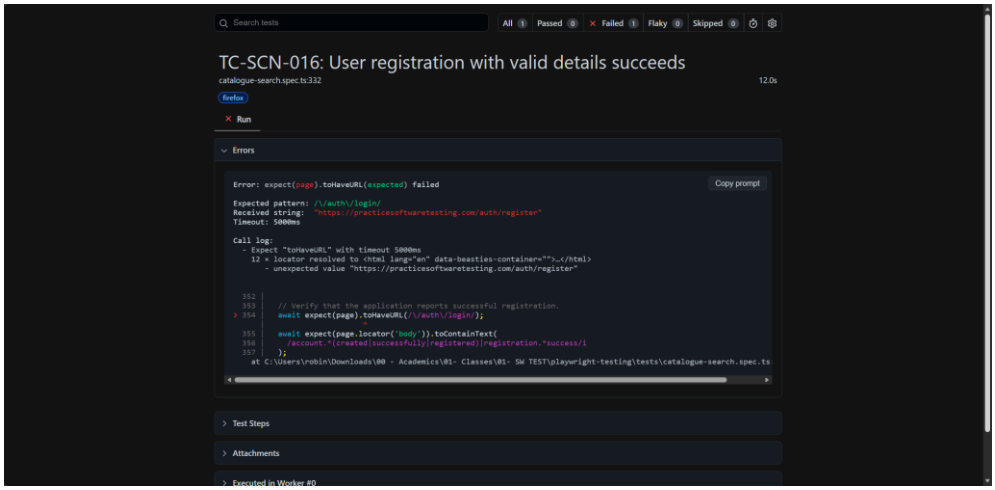
Figure 2 – Search or filter behavior

Suggested test: choose the relevant SRCH requirement scenario. Capture the search/filter input and resulting product list.



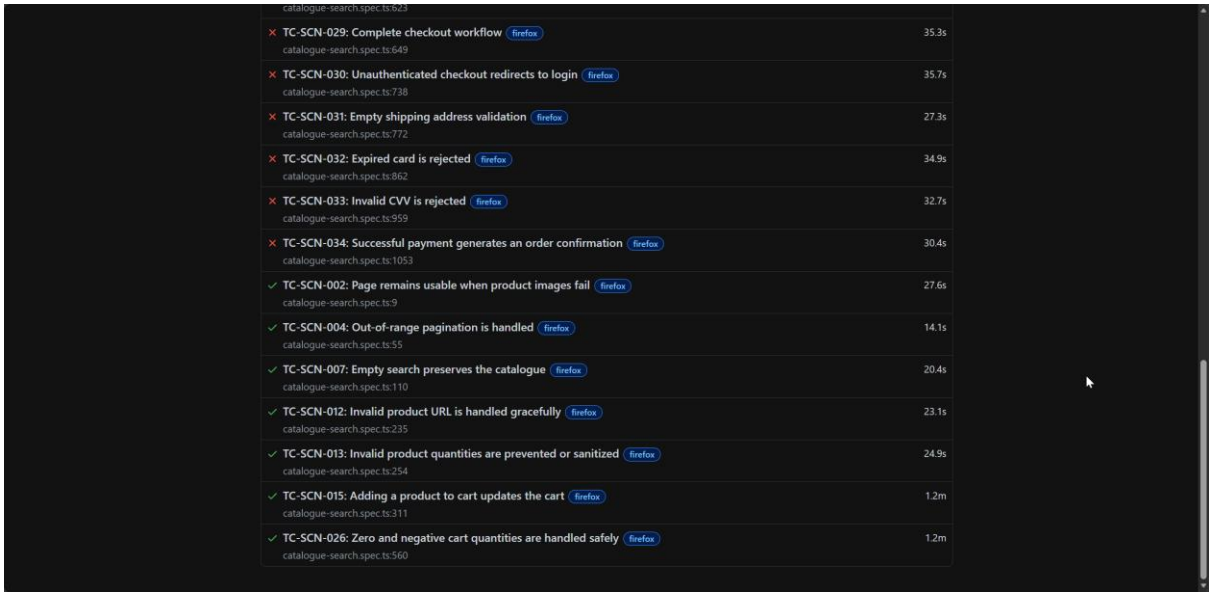
Playwright evidence for search/filter behavior ([TC-SCN-008]).

Figure 3 – Error while Testing in Playwright



Playwright evidence for erroneous test([TC-SCN-016])

Figure 4 – Test Case Suite Report in Playwright



Playwright evidence for Test Suite Report(Multiple tests were run concurrently therefore several failed which required User Inspection)

6. Execution Summary and Defect Log

Metric	Value	Notes
Total functional requirements	31	As listed in SRS Section 7
SRS-provided scenarios	34	TC-SCN-001 to TC-SCN-034
Executed test cases	34	All 34 Test Cases Ran
Passed / Failed / Blocked / Not Run	30 / 4 / 0 / 0	Based on the four failures recorded
Defects raised	4	As listed in Test Case Matrix section(failed TC's)